

# DBMS WEEK 4

## TOPIC: JOINS AND SET OPERATIONS

### STEP 1: CREATE TABLES

```
mysql> create database task4;
Query OK, 1 row affected (0.03 sec)

mysql> use task4;
Database changed
mysql> create table class (
  -> id int,
  -> name varchar(30)
  -> );
Query OK, 0 rows affected (0.06 sec)

mysql> desc class;
+-----+-----+-----+-----+-----+-----+
| Field | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| id    | int           | YES  |     | NULL    |       |
| name  | varchar(30)   | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
2 rows in set (0.01 sec)
```

### STEP 2: INSERT DATA

```
mysql> insert into class values
  -> (1,'abhi'),
  -> (2,'adam'),
  -> (4,'alex');
Query OK, 3 rows affected (0.03 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> select * from class;
+----+-----+
| id | name |
+----+-----+
|  1 | abhi |
|  2 | adam |
|  4 | alex |
+----+-----+
3 rows in set (0.00 sec)
```

### QUESTION :1 - CROSS JOIN

```
mysql> insert into class_info values
  -> (1,'delhi'),
  -> (2,'mumbai'),
  -> (3,'chennai');
Query OK, 3 rows affected (0.03 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> select * from class_info;
+----+-----+
| id | address |
+----+-----+
|  1 | delhi   |
|  2 | mumbai  |
|  3 | chennai |
+----+-----+
3 rows in set (0.00 sec)
```

### PART 2: INNER JOIN

```
mysql> create table class (  
-> id int,  
-> name varchar(30)  
-> );  
Query OK, 0 rows affected (0.06 sec)
```

```
mysql> desc class;
```

| Field | Type        | Null | Key | Default | Extra |
|-------|-------------|------|-----|---------|-------|
| id    | int         | YES  |     | NULL    |       |
| name  | varchar(30) | YES  |     | NULL    |       |

2 rows in set (0.01 sec)

```
mysql> create table class_info (  
-> id int,  
-> address varchar(30)  
-> );  
Query OK, 0 rows affected (0.06 sec)
```

```
mysql> desc class_info;
```

| Field   | Type        | Null | Key | Default | Extra |
|---------|-------------|------|-----|---------|-------|
| id      | int         | YES  |     | NULL    |       |
| address | varchar(30) | YES  |     | NULL    |       |

2 rows in set (0.00 sec)

```
mysql> insert into class_info values  
-> (1,'delhi'),  
-> (2,'mumbai'),  
-> (3,'chennai');  
Query OK, 3 rows affected (0.03 sec)  
Records: 3 Duplicates: 0 Warnings: 0
```

```
mysql> select * from class_info;
```

| id | address |
|----|---------|
| 1  | delhi   |
| 2  | mumbai  |
| 3  | chennai |

3 rows in set (0.00 sec)

```
mysql> insert into class values  
-> (1,'abhi'),  
-> (2,'adam'),  
-> (3,'alex'),  
-> (4,'anu');  
Query OK, 4 rows affected (0.03 sec)  
Records: 4 Duplicates: 0 Warnings: 0
```

```
mysql> select * from class;
```

| id | name |
|----|------|
| 1  | abhi |
| 2  | adam |
| 3  | alex |
| 4  | anu  |

4 rows in set (0.00 sec)

## QUESTION :2 - INNER JOIN

```
mysql> select * from class
-> inner join class_info
-> on class.id = class_info.id;
+-----+-----+-----+-----+
| id | name | id | address |
+-----+-----+-----+-----+
| 1 | abhi | 1 | delhi |
| 2 | adam | 2 | mumbai |
| 3 | alex | 3 | chennai |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

## QUESTION :3 - INNER JOIN

```
mysql> select class.name,
-> class_info.address
-> from class inner join class_info
-> on class.id = class_info.id;
+-----+-----+
| name | address |
+-----+-----+
| abhi | delhi |
| adam | mumbai |
| alex | chennai |
+-----+-----+
3 rows in set (0.00 sec)
```

## PART 3: NATURAL JOIN

## QUESTION: 4 - NATURAL JOIN

```
mysql> select*from class
-> natural join class_info;
+-----+-----+-----+
| id | name | address |
+-----+-----+-----+
| 1 | abhi | delhi |
| 2 | adam | mumbai |
| 3 | alex | chennai |
+-----+-----+-----+
3 rows in set (0.00 sec)
```

## PART 4: LEFT OUTER JOIN

```
mysql> select * from class;
+-----+-----+
| id | name |
+-----+-----+
| 1 | abhi |
| 2 | adam |
| 3 | alex |
| 4 | anu |
| 5 | ashish |
+-----+-----+
5 rows in set (0.00 sec)

mysql> select * from class_info;
+-----+-----+
| id | address |
+-----+-----+
| 1 | delhi |
| 2 | mumbai |
| 3 | chennai |
| 7 | noida |
| 8 | panipat |
+-----+-----+
5 rows in set (0.00 sec)
```

## QUESTION :5 - LEFT JOIN

```
mysql> select*from class
-> left outer join class_info
-> on class.id = class_info.id;
```

| id | name   | id   | address |
|----|--------|------|---------|
| 1  | abhi   | 1    | delhi   |
| 2  | adam   | 2    | mumbai  |
| 3  | alex   | 3    | chennai |
| 4  | anu    | NULL | NULL    |
| 5  | ashish | NULL | NULL    |

5 rows in set (0.02 sec)

QUESTION :6 - LEFT JOIN

```
mysql> select*from class
-> left join class_info
-> on class.id = class_info.id
-> where class_info.id is null;
```

| id | name   | id   | address |
|----|--------|------|---------|
| 4  | anu    | NULL | NULL    |
| 5  | ashish | NULL | NULL    |

2 rows in set (0.00 sec)

PART 5: RIGHT OUTER JOIN

QUESTION :7 - RIGHT JOIN

```
mysql> select*from class
-> right outer join class_info
-> on class.id = class_info.id;
```

| id   | name | id | address |
|------|------|----|---------|
| 1    | abhi | 1  | delhi   |
| 2    | adam | 2  | mumbai  |
| 3    | alex | 3  | chennai |
| NULL | NULL | 7  | noida   |
| NULL | NULL | 8  | panipat |

5 rows in set (0.00 sec)

QUESTION :8 - RIGHT JOIN

```
mysql> select*from class
-> right join class_info
-> on class.id = class_info.id
-> where class.id is null;
```

| id   | name | id | address |
|------|------|----|---------|
| NULL | NULL | 7  | noida   |
| NULL | NULL | 8  | panipat |

2 rows in set (0.00 sec)

## PART 6: FULL OUTER JOIN

### QUESTION :9 - FULL OUTER JOIN

```
mysql> select*from class
-> left join class_info
-> on class.id = class_info.id;
```

| id | name   | id   | address |
|----|--------|------|---------|
| 1  | abhi   | 1    | delhi   |
| 2  | adam   | 2    | mumbai  |
| 3  | alex   | 3    | chennai |
| 4  | anu    | NULL | NULL    |
| 5  | ashish | NULL | NULL    |

5 rows in set (0.00 sec)

### QUESTION :10 - FULL OUTER JOIN

```
mysql> select*from class
-> left join class_info
-> on class.id = class_info.id
-> where class_info.id is null
-> union
-> select*from class
-> right join class_info
-> on class.id = class_info.id
-> where class.id is null;
```

| id   | name   | id   | address |
|------|--------|------|---------|
| 4    | anu    | NULL | NULL    |
| 5    | ashish | NULL | NULL    |
| NULL | NULL   | 7    | noida   |
| NULL | NULL   | 8    | panipat |

4 rows in set (0.03 sec)

## PART 7: UNION

```
mysql> select * from first_table;
```

| id | ame  |
|----|------|
| 1  | abhi |
| 2  | adam |

2 rows in set (0.00 sec)

```
mysql> select * from second_table;
```

| id | ame     |
|----|---------|
| 2  | adam    |
| 3  | chester |

2 rows in set (0.02 sec)

### QUESTION :11 – UNION

```
mysql> select * from first_table
-> union
-> select * from second_table;
```

| id | ame     |
|----|---------|
| 1  | abhi    |
| 2  | adam    |
| 3  | chester |

3 rows in set (0.00 sec)

QUESTION :12 – UNION

```
mysql> select ame from first_table
-> union
-> select ame from second_table;
```

| ame     |
|---------|
| abhi    |
| adam    |
| chester |

3 rows in set (0.00 sec)

PART 8: UNION ALL

QUESTION :13 - UNION ALL

```
mysql> select * from first_table
-> union all
-> select * from second_table;
```

| id | ame     |
|----|---------|
| 1  | abhi    |
| 2  | adam    |
| 2  | adam    |
| 3  | chester |

4 rows in set (0.00 sec)

QUESTION :14 - UNION ALL

```
mysql> select count(*)
-> from
-> (
-> select * from first_table
-> union all
-> select * from second_table
-> ) a;
```

| count(*) |
|----------|
| 4        |

1 row in set (0.02 sec)

## PART 9: INTERSECT

### QUESTION :15 – INTERSECT

```
mysql> select f.*
-> from first_table f
-> inner join second_table s
-> on f.id = s.id
-> and f.ame = s.ame;
+-----+
| id | ame |
+-----+
| 2 | adam |
+-----+
1 row in set (0.02 sec)
```

### QUESTION :16 – INTERSECT

```
mysql> select f.ame
-> from first_table f
-> inner join second_table s
-> on f.ame = s.ame;
+-----+
| ame |
+-----+
| adam |
+-----+
1 row in set (0.00 sec)
```

## PART 10: MINUS

### QUESTION :17 – MINUS

```
mysql> select f.*
-> from first_table f
-> left join second_table s
-> on f.id = s.id
-> and f.ame = s.ame
-> where s.id is null;
+-----+
| id | ame |
+-----+
| 1 | abhi |
+-----+
1 row in set (0.00 sec)
```

### QUESTION :18 – MINUS

```
mysql> select f.ame
      -> from first_table f
      -> left join second_table s
      -> on f.ame = s.ame
      -> where s.id is null;
```

| ame  |
|------|
| abhi |

1 row in set (0.00 sec)

## ADVANCED INTERVIEW QUESTIONS

### QUESTION 19

```
mysql> select c.id, c.name, ci.address
      -> from class c
      -> inner join class_info ci
      -> on c.id = ci.id;
```

| id | name | address |
|----|------|---------|
| 1  | abhi | delhi   |
| 2  | adam | mumbai  |
| 3  | alex | chennai |

3 rows in set (0.00 sec)

### QUESTION 20

```
mysql> select c.id, c.name, case
      -> when ci.address is null
      -> then 'address missing'
      -> else 'address available'
      -> end as status
      -> from class c
      -> left join class_info ci
      -> on c.id = ci.id;
```

| id | name   | status            |
|----|--------|-------------------|
| 1  | abhi   | address available |
| 2  | adam   | address available |
| 3  | alex   | address available |
| 4  | anu    | address missing   |
| 5  | ashish | address missing   |

5 rows in set (0.02 sec)